vidssave.com Physics Revision 144P

Kinematics of the Mass

Determining Velocity from Acceleration

To find the velocity of the mass m at any time t, we use the relationship between acceleration a and velocity v'. Given that acceleration is the time derivative of velocity, we have $a = \frac{dv'}{dt}$. The problem states that the acceleration of the mass is $g - \alpha t$, where $\alpha = 1\ \mathrm{m/s^3}$.

By substituting the expression for acceleration, we get $\frac{dv'}{dt} = g - t$. To solve for v', we integrate both sides with respect to time: $\int dv' = \int (g - t) dt$. Given the initial condition that the mass starts from rest at $t = 0$ (where $v' = 0$), the integration yields $v' = gt - \frac{t^2}{2}$. This formula allows us to calculate the velocity of the mass at any instant t during its motion.

Dynamics of the Spring-Mass System

Equation of Motion

To analyze the forces acting on the mass m, we consider a free body diagram at an arbitrary time t. Let x represent the elongation of the spring beyond its natural length. The forces acting on the mass are the gravitational force mg acting downwards and the spring force kx acting upwards.

Applying Newton's second law, the net downward force is $mg - kx$, which must equal the mass times its acceleration. Using the given acceleration $a = g - t$, the equation of motion is $mg - kx = m(g - t)$. Simplifying this, the mg terms cancel out, leaving $-kx = -mt$, or $kx = mt$. This relationship defines the elongation of the spring as a function of time based on the mass's acceleration.

Relating Elongation to Velocity

To find the velocity v of the end P of the spring, we differentiate the equation $kx = mt$ with respect to time, yielding $k\frac{dx}{dt} = m$. Here, $\frac{dx}{dt}$ represents the rate at which the spring's elongation changes.

The rate of change of elongation, $\frac{dx}{dt}$, is equivalent to the relative velocity between the two ends of the spring. If v is the upward velocity of end P and v' is the downward velocity of the mass (end Q), the relative velocity is $v + v'$. Substituting this into our differentiated equation gives $k(v + v') = m$.

Solving for v, we get $v = \frac{m}{k} - v'$. Substituting the previously derived expression for v', we find the final velocity of end P as a function of time: $v = \frac{m}{k} - (gt - \frac{t^2}{2})$, which simplifies to $v =$

$\frac{m}{k} - gt + \frac{t^2}{2}$.

Mechanical System Analysis: Rods and Hinges

Geometric Configuration and Locus of Motion

The system consists of two rods, P and Q. Rod P has a length of $1\ \mathrm{m}$ and is hinged at a fixed point A. At the free end of rod P, there is a ring labeled C. Rod Q is a longer rod that is hinged at a fixed point B and passes through the ring at C. Both rods are constrained to move within the same plane.

Because rod P is hinged at A and has a fixed length of $1\ \mathrm{m}$, the point C is constrained to move along a circular path. The center of this circle is the hinge point A, and the radius of the circle is $1\ \mathrm{m}$. This defines the locus of point C as a circle centered at A with a radius of $1\ \mathrm{m}$.

Determining the Distance Between Hinges

The problem requires finding the distance between the two fixed hinge points, A and B, based on the variation of the angle ϕ (the angle of rod Q) with respect to the angle θ (the angle of rod P). As both rods swivel, the position of C changes, and for a given angle ϕ, there may be two possible positions for C, corresponding to two different values of θ.

The angle ϕ reaches its maximum value when rod Q becomes tangent to the circular path of point C. At this point of tangency, the geometry forms a right-angled triangle where the hypotenuse is the distance between the hinges, AB. The side adjacent to the angle θ is the radius of the circle, which is $1\ \mathrm{m}$.

Given that the maximum value of ϕ occurs when $\theta = 60^\circ$, we can use trigonometry to solve for the distance AB. In the right triangle formed by the hinges and the point of tangency, the relationship is defined by the secant function: $AB \cdot \cos(60^\circ) = 1\ \mathrm{m}$.

Rearranging this formula gives the distance AB:

$$AB = \frac{1}{\cos(60^\circ)} = 1 \cdot \sec(60^\circ)$$

Since $\sec(60^\circ) = 2$, the distance between the hinges A and B is 2 meters.

Calculating the Center of Mass of a Composite System

Conceptual Decomposition of the System

To find the center of mass of a tank containing a liquid of mass 5 kg and a submerged sphere of mass 2 kg (with density $2ho$), it is helpful to decompose the sphere into two simpler components.

If the sphere's volume were occupied by the same liquid (density ρ), its mass would be 1 kg. Since the actual sphere has a density of $2ho$, its mass is 2 kg.

We can treat the sphere as a combination of a 1 kg volume of liquid (density ρ) and a point mass of 1 kg located at the center of the sphere. By replacing the sphere with this liquid volume and the point mass, the entire tank effectively becomes a uniform liquid of 6 kg (the original 5 kg plus the 1 kg added to fill the sphere's volume) plus a discrete 1 kg point mass located at the sphere's original center, C.

Calculating the Height of the Center of Mass

With the system simplified into a uniform liquid and a point mass, we can calculate the height of the center of mass above the base P. Let $M_1 = 6\ \text{kg}$ be the mass of the liquid, with its center of mass at $H_1 = 2\ \text{m}$ above P (since the total liquid height is 4 m). Let $M_2 = 1\ \text{kg}$ be the point mass at C, located at $H_2 = 3\ \text{m}$ above P.

The formula for the center of mass height H_{cm} is given by:

$$H_{cm} = \frac{M_1 H_1 + M_2 H_2}{M_1 + M_2}$$

Substituting the known values:

$$H_{cm} = \frac{(6\ \text{kg} \times 2\ \text{m}) + (1\ \text{kg} \times 3\ \text{m})}{6\ \text{kg} + 1\ \text{kg}}$$

$$H_{cm} = \frac{12 + 3}{7} = \frac{15}{7}\ \text{meters}$$

This calculation provides the height of the center of mass for the entire system relative to the bottom of the tank P.

Problem Setup and Non-Inertial Frame Analysis

System Dynamics and Frame of Reference

The system consists of a particle of mass m constrained within a circular tube mounted on a trolley. Initially, the trolley is in motion. When the brakes are applied, the trolley undergoes a deceleration a directed to the left. This deceleration is equivalent to an acceleration a directed to the left.

To analyze the motion of the particle, it is effective to use a non-inertial frame of reference attached to the trolley. In this frame, the particle experiences standard forces like gravity (mg acting downwards) and the normal reaction force (N) from the tube. Additionally, because the frame is accelerating, a pseudo force must be included in the analysis.

The pseudo force acting on the particle is defined as ma, directed opposite to the trolley's acceleration. Since the trolley accelerates to the left, the pseudo force acts on the particle to the right.

Work-Energy Principle Application

Calculating Work Done by Forces

The work-energy principle is applied to the particle as it moves from its initial position A to the top of the tube B. The total work done is the sum of the work done by gravity, the normal reaction, and the pseudo force.

Gravity acts downwards throughout the motion. Given the vertical displacement of the particle is $2r$ (the diameter of the circular tube), the work done by gravity is $-mg(2r)$.

The normal reaction force N acts perpendicular to the surface of the tube at all times. Since the particle's displacement is always tangential to the tube, the normal force is always perpendicular to the displacement. Consequently, the work done by the normal reaction is zero.

The pseudo force ma acts horizontally to the right. This force only acts while the trolley is decelerating. The displacement of the particle in the direction of the pseudo force is $r \sin\theta$, where θ is the angle of swivel. Thus, the work done by the pseudo force is $ma(r \sin\theta)$. Once the trolley comes to rest, the pseudo force ceases to act, contributing no further work for the remainder of the journey from E to B.

Kinetic Energy in Non-Inertial Frames

Work-Energy Theorem in the Trolley Frame

In a non-inertial frame, such as a trolley undergoing acceleration, the work-energy theorem must account for pseudo forces. The change in kinetic energy of a particle within this frame is equal to the total work done by all forces, including gravity and the pseudo force acting on the particle.

In the provided example, a particle starts at rest relative to the trolley at point A and reaches point B, where it is again at rest relative to the trolley. Because the initial and final velocities in the trolley's frame are both zero, the total change in kinetic energy is $0 - 0 = 0$.

By setting the sum of the work done by gravity and the pseudo force to zero, one can solve for the acceleration a of the trolley. The derived relationship for this specific system is $a = \frac{2g}{\sin\theta}$.

Thermodynamics of Dissociation

Conservation of Internal Energy in Adiabatic Processes

In an adiabatic process within a rigid container, the total internal energy of the system is conserved. When a chemical reaction, such as the dissociation of hydrogen molecules (H_2) into hydrogen atoms (H), occurs, the energy balance must account for both the change in thermal energy and the dissociation energy required to break the molecular bonds.

The initial internal energy is calculated using the number of moles n, the temperature T, and the molar heat capacity at constant volume C_v. For a diatomic gas like H_2, $C_v = \frac{5}{2}R$. Given $R = 2\ \text{cal/mole-K}$, $C_v = 5\ \text{cal/mole-K}$. The initial energy is $n \cdot C_v \cdot T_{initial} = 4 \cdot 5 \cdot 500 = 10,000\ \text{calories}$.

After dissociation, the system contains 2 moles of monoatomic hydrogen ($C_v = 3$) and 3 moles of remaining H_2 ($C_v = 5$) at a new temperature $100T$. The energy balance equation is:

$$500 \cdot 4 \cdot 5 = (2 \cdot 3 \cdot 100T) + (3 \cdot 5 \cdot 100T) + 2000$$

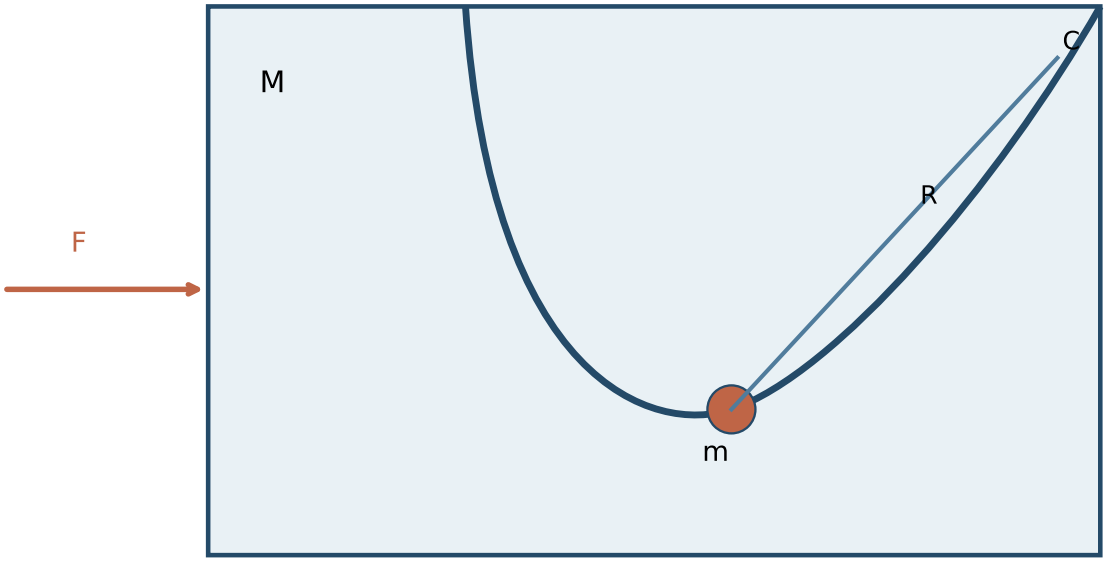
Simplifying this yields $10,000 = 600T + 1500T + 2000$, which reduces to $21T = 80$. Solving for T gives $T = \frac{80}{21} \approx 3.8$. The closest integer value for T is 4.

Statics of a Wedge with a Moving Particle

Maintaining Equilibrium with Variable Force

When an insect of mass m crawls at a constant speed v along a circular quadrant of radius R on a wedge of mass M, the wedge experiences dynamic forces. To keep the wedge stationary on a smooth surface, an external variable force F must be applied.

The force F must counteract the horizontal component of the force exerted by the insect on the wedge. As the insect moves, the angle θ of its position relative to the horizontal changes, necessitating a variable force to maintain equilibrium at every instant.



Physics Mechanics Diagram: Block and Curved Surface

Dynamics of a Crawling Insect on a Wedge

System Analysis and Newton's Second Law

To determine the relationship between the force F applied to a wedge, the constant speed v of an insect crawling on it, and the angle $heta$, we analyze the system as a whole. The insect crawls along a quadrant of radius r. At any angle $heta$ with respect to the horizontal, the insect experiences a centripetal force directed toward the center of the circular path, given by $F_c = \frac{mv^2}{r}$, where m is the mass of the insect.

Because this centripetal force vector makes an angle $heta$ with the horizontal, its horizontal component is $\frac{mv^2}{r}\cos\theta$. By treating the wedge (mass M) and the insect (mass m) as a single system, we can apply Newton's second law in the horizontal direction. Since the wedge is stationary, its horizontal acceleration is zero. The net horizontal force F applied to the system must equal the sum of the horizontal mass-acceleration products of the components:

$$F = M(0) + m\left(\frac{v^2}{r}\cos\theta\right)$$

This simplifies to the relationship:

$$F = \frac{mv^2}{r}\cos\theta$$

This approach is efficient because it avoids calculating the internal normal reaction force between the insect and the wedge, which is not required to solve for the external force F.

Strategic Simplification

The key to solving this problem effectively is the choice of the system boundary. By defining the system to include both the wedge and the insect, the normal force between them becomes an internal force. Internal forces cancel out when applying Newton's second law to the entire system, significantly simplifying the derivation. This method is preferred when the internal interactions are not the focus of the inquiry.

Fluid Dynamics in a Rotating U-Tube

Calculating Height Difference in a Rotating System

When a U-tube filled with liquid is rotated about a vertical axis with angular velocity ω, the liquid levels in the two limbs change due to the centrifugal effect. The difference in height h between the free surfaces of the liquid in the two limbs is determined by the rotation. Given the geometry of the tube, h is calculated by subtracting the height of the liquid in one limb from the other. In this example, with heights of 25 cm and 20 cm, the difference is $h = 5 \text{ cm} = 0.05 \text{ m}$.

The relationship between the height difference h, the angular velocity ω, the radius r of the rotation, and the acceleration due to gravity g is given by the standard fluid mechanics formula:

$$h = \frac{\omega^2 r^2}{2g}$$

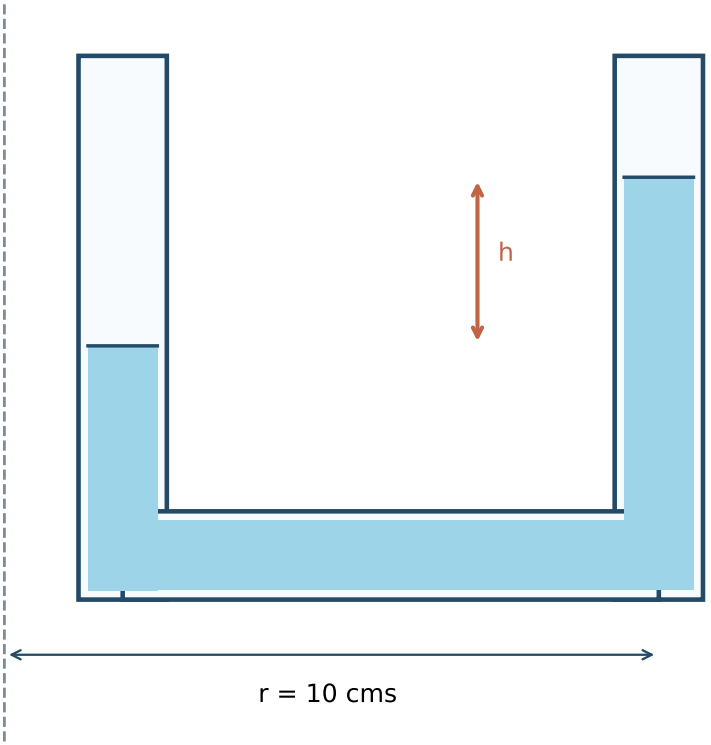
This formula is derived from the pressure distribution in a rotating fluid, where the pressure gradient balances the centrifugal force. To find ω, we rearrange the equation:

$$\omega = \sqrt{\frac{2gh}{r^2}}$$

Using $g = 10 \text{ m/s}^2$, $h = 0.05 \text{ m}$, and $r = 0.1 \text{ m}$, we substitute these values into the equation:

$$\omega = \sqrt{\frac{2 \cdot 10 \cdot 0.05}{0.1^2}} = \sqrt{\frac{1}{0.01}} = \sqrt{100} = 10 \text{ rad/s}$$

This result demonstrates the direct application of rotational fluid mechanics to determine the angular speed required to maintain a specific liquid column displacement.



Rotating U-tube system

Free Surface of a Rotating Fluid

Parabolic Trajectory and Height-Radius Relationship

When a fluid is rotated, the free surface assumes a parabolic shape. The relationship between the height h of the surface at a distance r from the axis of rotation and the angular velocity ω is given by the formula:

$$h = \frac{\omega^2 r^2}{2g}$$

where g is the acceleration due to gravity.

This formula is derived from the balance of forces acting on a fluid element at the surface, where the pressure gradient must support the weight of the fluid and provide the necessary centripetal acceleration. Understanding this derivation is essential for solving variations of the problem, such as determining the angular velocity required to achieve a specific height at a given radius.

Geometrical Optics: Mirror and Wall Displacement

System Setup and Initial Reflection

Consider a horizontal mirror and a light source T that directs a beam onto the mirror at an angle of incidence $\theta = 37^\circ$ with the vertical. The reflected ray strikes a vertical wall at a spot S_1. Because the angle of incidence equals the angle of reflection, the reflected ray also makes an angle θ with the vertical.

When the mirror is shifted upward by a distance of $1\ \text{cm}$, the point of incidence on the mirror changes from P to Q. This shift in the mirror's position causes the reflected ray to strike the wall at a new position. If the wall remained stationary, the spot would shift vertically due to the change in the path of the reflected ray.

Calculating Vertical Shift with Moving Boundaries

To calculate the total vertical shift when the mirror moves up by $1\ \text{cm}$ and the wall moves to the right by $1\ \text{cm}$, we analyze the geometry of the reflected rays. By constructing triangles based on the points of incidence and the wall positions, we can identify congruent triangles. Specifically, the geometry dictates that the vertical shift caused by the mirror's movement is related to the distance the mirror moved and the angle θ.

The total vertical shift is the sum of the shift caused by the mirror's displacement and the additional shift caused by the wall's movement. The initial vertical shift (if the wall were stationary) is $2\ \text{cm}$ based on the parallelogram formed by the ray paths. The additional vertical shift h caused by moving the wall $1\ \text{cm}$ to the right is calculated using trigonometry:

$$h = 1 \cdot \cot(\theta)$$

Given $\theta = 37^\circ$, we use $\cot(37^\circ) = \frac{\cos(37^\circ)}{\sin(37^\circ)} = \frac{4}{3}$. Thus, the additional shift is $\frac{4}{3}\ \text{cm}$, and the net vertical shift is the sum of the initial $2\ \text{cm}$ and this additional component.

Vertical Shift Calculation

Calculating Net Vertical Shift

The net vertical shift of a light ray is determined by the compounded effects of moving components, such as a mirror and a wall. In this scenario, the total vertical shift (VS) is the sum of the displacement caused by the mirror and the displacement caused by the wall.

Given a mirror displacement of $2\ \text{cm}$ and a wall displacement calculated as $4/3\ \text{cm}$ (derived from $h = 1 \cdot \cot(37^\circ)$), the total shift is calculated as follows:

$$VS = 2 + \frac{4}{3} = \frac{6+4}{3} = \frac{10}{3}\ \text{cm}$$

This result represents the final vertical displacement of the light ray on the target surface.

Two-Lens Optical System

Deviation and Focal Length Relationship

When a ray of light passes through a combination of two lenses with focal lengths f_1 and f_2, it undergoes deviation. For paraxial rays, the angle of deviation δ produced by a lens is approximately given by the ratio of the height of incidence h to the focal length f:

$$\tan \delta \approx \delta = \frac{h}{f}$$

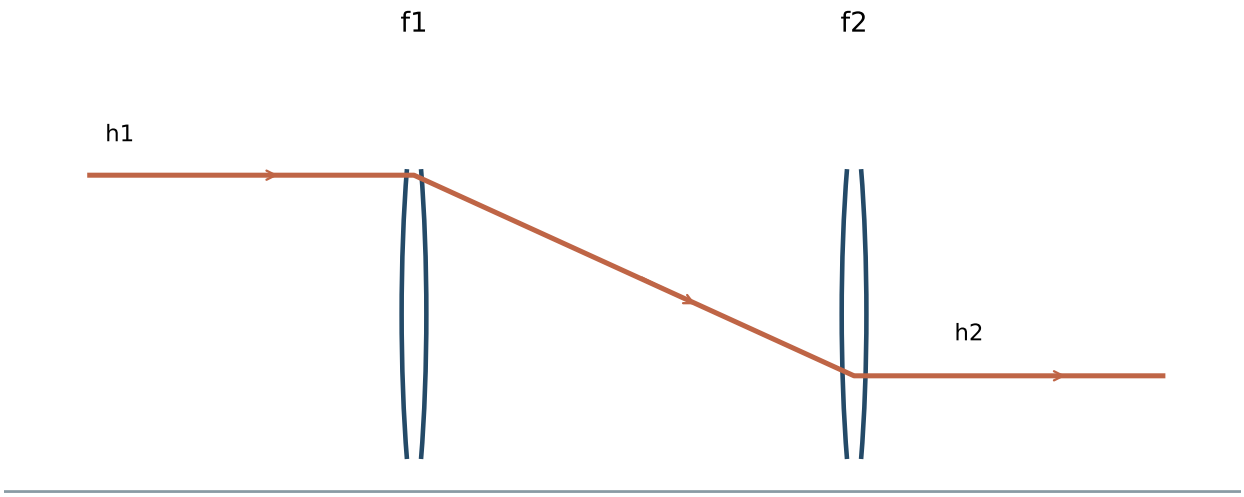
Here, h is the distance of the ray from the principal axis at the point of incidence. If a ray is incident on two lenses and experiences the same deviation δ, we can equate the expressions for each lens:

$$\delta = \frac{h_1}{f_1} = \frac{h_2}{f_2}$$

Rearranging this equality allows us to compare the focal lengths based on the heights of incidence:

$$\frac{f_1}{f_2} = \frac{h_1}{h_2}$$

If the height of incidence on the first lens (h_1) is greater than the height on the second lens (h_2), then the ratio f_1/f_2 must be greater than 1, implying $f_1 > f_2$.



Two-Lens Optical System Ray Diagram

LC Circuit Oscillations

Determining Time Period of Oscillation

In an LC circuit, the energy stored in the capacitor is equal to the energy stored in the inductor at any given instant. This relationship is expressed as:

$$\frac{Q^2}{2C} = \frac{1}{2}LI^2$$

where Q is the charge, C is capacitance, L is inductance, and I is current. The angular frequency of oscillation for an LC circuit is defined as $\omega = 1/\sqrt{LC}$.

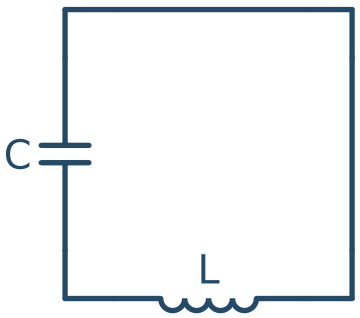
To find the time period T, we first solve for the product LC using the energy equation:

$$LC = \frac{Q^2}{I^2}$$

Substituting this into the formula for the time period $T = 2\pi/\omega = 2\pi\sqrt{LC}$, we get:

$$T = 2\pi\sqrt{\frac{Q^2}{I^2}} = 2\pi\frac{Q}{I}$$

This formula provides the time period of the electrical oscillations in terms of the instantaneous charge and current.



LC Circuit

Thermal Expansion

Calculating Change in Length

The coefficient of linear expansion α is defined by the fractional change in length per unit change in temperature:

$$\alpha = \frac{1}{L}\frac{dL}{dT}$$

To find the total change in length for a given temperature change ΔT, we rearrange the equation to isolate the differential terms:

$$\frac{dL}{L} = \alpha\, dT$$

Integrating both sides of this equation allows us to determine the relationship between the initial length L_1 and the final length L_2 over a temperature range:

$$\int \frac{dL}{L} = \int \alpha\, dT$$

$$\ln\left(\frac{L_2}{L_1}\right) = \alpha \Delta T$$

This logarithmic relationship describes how the length of the rod evolves as the temperature changes.

Thermal Expansion

Linear Expansion Coefficient

The coefficient of linear expansion, denoted by α, describes the fractional change in length of a material per unit change in temperature. It is defined by the relationship $\alpha = \frac{1}{L}\frac{dL}{dT}$. By integrating this expression, we arrive at the logarithmic relationship $\ln(\frac{L_2}{L_1}) = \alpha \Delta T$.

In a practical application where $\alpha = 10^{-3}/^\circ C$ and $\Delta T = 1000K$, the product $\alpha \Delta T$ equals 1. Substituting this into the integrated formula yields $\ln(\frac{L_2}{L_1}) = 1$. Since the base of the natural logarithm is $e \approx 2.7$, it follows that $\frac{L_2}{L_1} = e$, or $L_2 \approx 2.7 L_1$. This demonstrates that the final length is significantly greater than the initial length.

Fluid Dynamics in Gravity-Free Space

Bernoulli's Principle and Force-Velocity Relationship

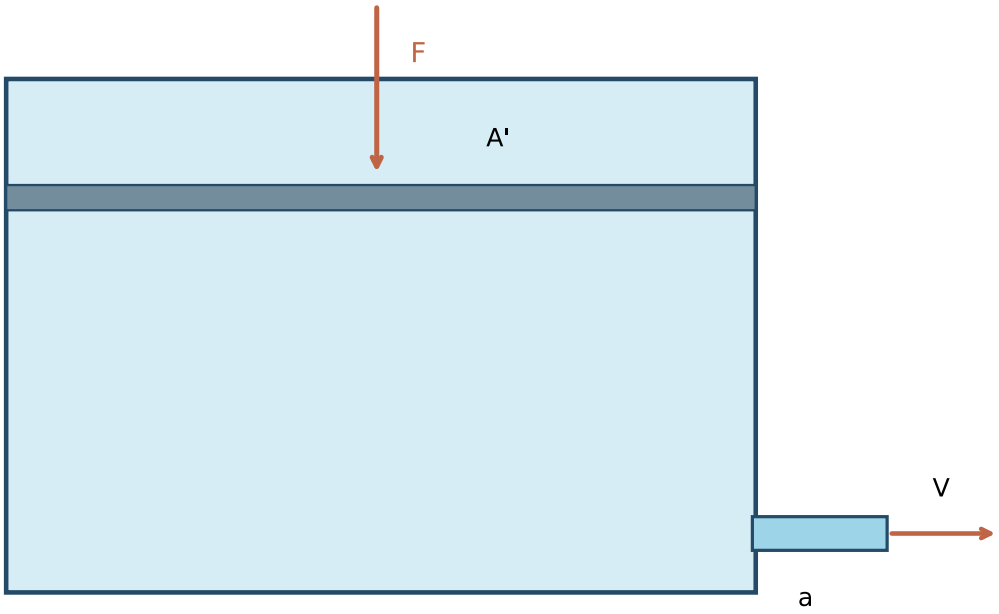
Consider a piston-driven injection system in a gravity-free environment. A force F is applied to a piston with cross-sectional area A', forcing liquid out of an orifice with area A at velocity v. Applying Bernoulli's theorem between the piston surface and the exit orifice, and noting that both are at atmospheric pressure, the pressure difference created by the force is $\frac{F}{A'}$.

The energy balance equation is $\frac{F}{A'} + \frac{1}{2}\rho v'^2 = \frac{1}{2}\rho v^2$, where v' is the velocity of the piston. From the continuity equation, $A'v' = Av$. Given that the orifice area A is much smaller than the piston area A', it follows that $v \gg v'$. Consequently, the kinetic energy term associated with the piston ($\frac{1}{2}\rho v'^2$) is negligible. This simplifies the equation to $\frac{F}{A'} \approx \frac{1}{2}\rho v^2$, which shows that the force F is proportional to the square of the efflux velocity, $F \propto v^2$.

Power Requirements

Power P is defined as the product of force and velocity, specifically $P = F \cdot v'$. Using the previously derived relationship where $F \propto v^2$ and the continuity relation $v' = \frac{Av}{A'}$, we can substitute these into the power equation.

Since v' is directly proportional to v (given constant areas A and A'), the power expression becomes $P \propto (v^2) \cdot (v)$, which simplifies to $P \propto v^3$. Thus, the power required to maintain the efflux velocity is proportional to the cube of that velocity.



Fluid dynamics container schematic

Fluid Dynamics and Proportionality

Proportionality of Power, Time, and Work

In a fluid system where a tank is being emptied, the power required is proportional to the cube of the velocity ($P \propto v^3$). This relationship arises from the definition of power as the product of force and velocity ($P = F \cdot v$). Since the force required to maintain the flow is proportional to the square of the velocity ($F \propto v^2$), the power scales as $v^2 \cdot v = v^3$.

The time (T) required to empty a tank is determined by the total volume of the tank divided by the volume flow rate. Given the flow rate is the product of area and velocity ($A \cdot v$), the time is inversely proportional to the velocity ($T \propto v^{-1}$).

The work done (WD) to empty the tank is calculated as the product of the force and the displacement ($WD = F \cdot S$). Because the force F is proportional to v^2, the work done is also proportional to the square of the velocity ($WD \propto v^2$).

Electrostatics and Potential Functions

Deriving Potential from a Variable Electric Field

Given a variable electric field in the xy-plane defined as $\vec{E} = x\hat{i} + y\hat{j}$, we can determine the electric potential $V(x, y)$ by considering the work done by the field. When a particle moves from

(x, y) to $(x + dx, y + dy)$, the work done by the electric field is $\vec{E} \cdot d\vec{r}$, where $d\vec{r} = dx\hat{i} + dy\hat{j}$.

By definition, the work done by the electric field is equal to the negative change in potential: $dW = -dV$. Substituting the field components, we get $-dV = (x\hat{i} + y\hat{j}) \cdot (dx\hat{i} + dy\hat{j}) = x dx + y dy$.

This expression can be simplified using the algebra of differentials: $x dx + y dy = \frac{1}{2} d(x^2 + y^2)$. Integrating both sides yields $-V + C = \frac{1}{2}(x^2 + y^2)$, where C is the constant of integration.

Applying Boundary Conditions

To find the specific potential function, we use the given boundary condition: at the point $(2, 2)$, the potential V is 0. Substituting these values into the derived equation $-V + C = \frac{1}{2}(x^2 + y^2)$ allows us to solve for C.

Plugging in $x = 2, y = 2$, and $V = 0$ gives: $-0 + C = \frac{1}{2}(2^2 + 2^2) = \frac{1}{2}(4 + 4) = 4$. Thus, $C = 4$.

The final potential function is $4 - V = \frac{1}{2}(x^2 + y^2)$, which can be rearranged as $2(4 - V) = x^2 + y^2$. This equation describes the relationship between the potential and the spatial coordinates.

Derivation of Equipotential Surfaces

Mathematical Derivation of Potential

The electric potential V is derived from the electric field $\vec{E} = x\hat{i} + y\hat{j}$. Using the relationship $-dV = \vec{E} \cdot d\vec{r}$, where $d\vec{r} = dx\hat{i} + dy\hat{j}$, we obtain $-dV = x dx + y dy$. This expression is the differential of the function $\frac{1}{2}(x^2 + y^2)$. Integrating this leads to the equation $-V + C = \frac{1}{2}(x^2 + y^2)$, which can be rearranged to the form $x^2 + y^2 = 2(4 - V)$ given specific boundary conditions where $V(2, 2) = 0$.

Geometry of Equipotential Curves

For any constant value of V, the equation $x^2 + y^2 = \text{constant}$ describes a circle centered at the origin $(0, 0)$. These equipotential curves are therefore concentric circles. The assertion that these circles are centered at $(2, 2)$ is incorrect; the center is the origin $(0, 0)$.

Constraints on Potential

The potential V is constrained by the physical requirement that $x^2 + y^2$ must be non-negative. From the equation $x^2 + y^2 = 2(4 - V)$, if $V > 4$, the right-hand side becomes negative, which is impossible for real coordinates (x, y). Therefore, the potential at any point cannot exceed 4 units.

Work Done on Equipotential Surfaces

The work done in moving a test charge Q slowly between two points A and B is given by $W = Q(V_B - V_A)$. If points A and B lie on the same equipotential curve, then $V_A = V_B$, and the potential difference is zero. Consequently, the work done in moving a charge between any two points on the same concentric circle is zero.

Relationship Between Radius and Potential

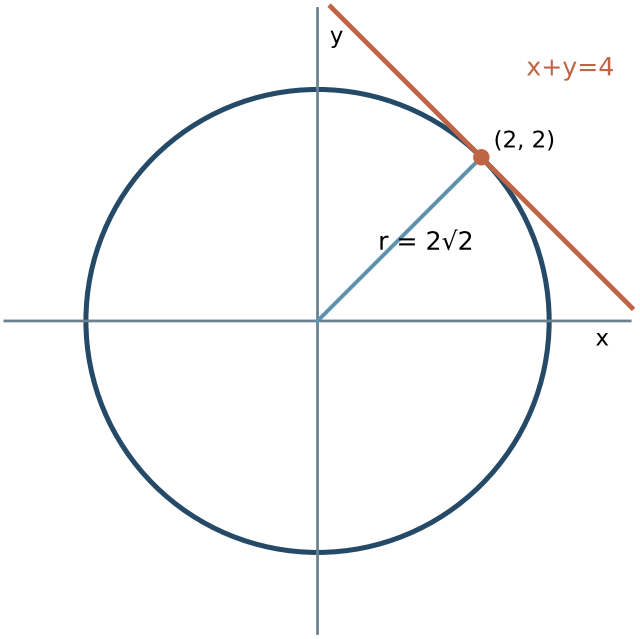
The radius R of an equipotential circle is given by $R = \sqrt{2(4 - V)}$. This relationship shows that as the potential V increases, the radius R decreases. Conversely, as the radius of the concentric circles increases, the potential V decreases. This implies that moving outward from the origin toward larger circles corresponds to moving toward lower potential values.

Electrostatic Potential and Geometry

Equipotential Lines and Circles

For a circle centered at (0,0) passing through (2,2), radius r = 2√2. If this is V=0, lines x+y=c where c>4 represent larger radii.

As radius r increases, potential V decreases. Thus, points on lines where c>4 correspond to negative potential values.



Circle and Tangent Line Geometry

Potential and Radius Relationship

Inverse relationship: as radius increases, potential decreases. Circles with r > 2√2 have negative potential.

Points on line x+y=c (c>4) lie on circles larger than the zero-potential boundary, confirming negative potential.

Voltmeter Sensitivity

Deriving Voltage Sensitivity

Voltmeter uses series resistance R with galvanometer. Current Ig = ΔV / (Rg + R).

Deflection θ is proportional to current: (NBA)Ig = Cθ.

Sensitivity S = θ/ΔV = NBA / [C(Rg + R)].

Adjusting Sensitivity

To decrease sensitivity S, increase denominator (R, Rg, or C) or decrease numerator (N, B, or A).

Increasing series resistance R or torsional constant C reduces sensitivity.

Voltage Sensitivity of a DC Voltmeter

Derivation of Sensitivity Formula

The voltage sensitivity of a DC voltmeter is defined as the ratio of the deflection produced, θ, to the potential difference applied, ΔV. To derive this, we begin with the fundamental relationship for a galvanometer current, I_g, when connected in series with a resistance R to measure a voltage ΔV: $I_g = \frac{\Delta V}{R_g+R}$, where R_g is the internal resistance of the galvanometer.

The torque produced by the current in the galvanometer coil is given by $(NBA)I_g = C\theta$, where N is the number of turns, B is the magnetic field, A is the area of the coil, and C is the torsional stiffness constant of the suspension. By substituting the expression for I_g into the torque equation, we obtain $(NBA)\left(\frac{\Delta V}{R_g+R}\right) = C\theta$.

Rearranging this equation to solve for the sensitivity, $\frac{\theta}{\Delta V}$, yields the final expression: $Sensitivity = \frac{\theta}{\Delta V} = \left(\frac{NBA}{C}\right)\left(\frac{1}{R_g+R}\right)$. This formula demonstrates that sensitivity is directly proportional to the number of turns N, the magnetic field B, and the area A, and inversely proportional to the torsional constant C and the total resistance $(R_g + R)$.

Factors Affecting Sensitivity

Based on the derived formula, the voltage sensitivity can be decreased by manipulating the variables in the expression. Since sensitivity is proportional to N, decreasing the number of turns N will result in a decrease in the overall sensitivity value. Similarly, increasing the total resistance $(R_g + R)$ will also decrease the sensitivity.

Understanding these relationships is essential for the practical conversion of a galvanometer into a voltmeter. Students are encouraged to master these basic rudiments, as they represent fundamental concepts in electrical instrumentation that frequently appear in examinations. Recognizing these relationships allows one to easily identify how design changes impact the performance of the measuring device.